Related Work: GNN-based EMG Gesture Recognition (Group 05, NinaPro DB2)

Title	Authors	Year	Dataset	Application area	Method / Model	Metrics	Strengths	Limitations	Research gap	Relation to our work
Cross-User Electromyography Pattern Recognition Based on a Novel Spatial-Temporal Graph Convolutional Network (IEEE TNSRE)	M. Xu, X. Chen, Y. Ruan, X. Zhang	2024	Own HD-sEMG database: 128 channels, 11 subjects, 17 gestures, 28,424 samples	Myoelectric control / prosthesis; cross-user gesture recognition	CNN-MSTGCN: CNN feature extractor + multi-view (distance graph and adaptive functional graph) spatial-temporal GCN (Chebyshev) with spatial and temporal attention; transfer learning (TL-CMSTGCN)	Recognition rate. Within-subject 96.9%; user-independent 68.0% (ResNet50 47.5%, LSTM 57.1%); with TL 92.3%; inference 2.61 ms/sample	Learns functional links between non-adjacent muscles; ablation shows each module helps; beats ResNet50/LSTM; transfer learning cuts user training burden	User-independent accuracy (68%) still low; offline only; private dataset; needs fine-tuning data for 92.3%	Cross-user generalisation without per-user calibration is unsolved; graph choice not compared systematically	Shows GNNs can beat CNN/LSTM across users. We will test graph variants on a public dataset (Nina Pro DB2) with a subject-independent split
Decoding	H. Lee, M.	2025	5 public	Prosthetic contro	GCN (message	Accuracy, intra-s	Many datasets;	Random 8:2	Graph structure is	Closest templ

Gestures in Electromyography: Spatio temporal Graph Neural Networks for Generalizable and Interpretable Classification (IEEE TNSRE)	Jiang, J. Yang, Z. Yang, Q. Zhao		datasets: NinaPro DB5, Capg Myo A and B, UCI EMG, Band Myo	l; hand gesture classification; explainable AI; sensor reduction	passing) on explicit graphs built from sensor layout: KNN-1, KNN-2, fully connected, each with or without sensor-wise temporal (SW) edges; also GAT, STGCN, Graph LSTM, GCLSTM	ubject and inter-subject. Intra: over 99.3% on all datasets with FC+SW or KNN-2+SW; inter-subject up to 99.4% (Band Myo), 92.4% (DB5), 93.7% (Capg Myo B); sensor-importance maps	compares six graph structures; interpretable sensor importance; stays accurate with 28 of 128 sensors	split over windows (possible leakage between near-identical windows); dense graphs are costly; no real-time test; NinaPro DB2 not used	fixed by layout; no study on sparse 12-channel setups such as DB2; evaluation protocol may be optimistic	ate for our graph-construction comparison (KNN vs fully connected vs temporal edges). We apply it to DB2 with a repetition-wise split
A Novel Multi-Feature Fusion Network With	F. Hu, M. Qian, K. He, W.-A. Zhang, X. Yang	2024	Own MyoData (17 subjects, 11 gestures, Myo armba	Armband gesture recognition for prosthetics and	MFN-SPSCA: Pearson-correlation graph with 4 partitioning	Accuracy, precision, recall, F1, Kappa, AUC, inference	Fuses time, space and frequency features; best DB5 result	Subject-dependent only; authors state it generalises poorly	No cross-subject evaluation; complexity makes it	Gives a correlation-based graph option and DB5 baselines.

Spatial Partitioning Strategy and Cross-Attention for Armband-Based Gesture Recognition (IEEE TNSRE)			nd 8 sEMG + 3 ACC) and NinaPro DB5 (10 subjects, 53 gestures, 2 Myo)	exoskeletons	strategies in a 9-layer STGCN, plus multi-scale CNN-BiLSTM, CWT branch and cross-attention fusion	time. 98.78 % (Myo Data), 91.60 % (DB5); 12.18 ms and 21.25 ms per sample	s among compared methods; ablations and t-SNE/CAM visualisation; fast enough for real time	to unseen users; large multi-branch model; sEMG+ACC needs IMU	hard to tell what the graph itself contributes	We keep the model simple to isolate the effect of the graph on raw sEMG
Hand Gesture Recognition From Surface Electromyography Signals With Graph Convolutional Network and Attention Mechanisms (IEEE	H. Zhou, H. T. Le, S. Zhang, S. L. Phung, G. Alici	2025	NinaPro DB2 (40 intact subjects, 12 Delsys sensors, Exercises B/C/D, 49 gestures) and own UOW dataset (10 subjects, 8-ch armba	Prosthetic hand control; robustness to limb position	CovGCN: shared prioring graph refined dynamically by a covariance-based topology refinement module (CovTRM); graph-conv blocks with multi-	Classification accuracy. DB2 Ex. B 88.02 %, C 81.30 %, D 93.30 %; UOW 84.69 % vs CNN 81.98 %; 1.07M parameters, 29.6 ms/signal	Uses NinaPro DB2 raw sEMG directly; dynamic graph; beats CNNs and classical ML; light model; limb-position test	Intact subjects only; offline; subject-specific, no cross-subject test; slightly below a Transformer on Ex. B (88.91 %)	First dynamic-graph model on sEMG, but no unseen-subject evaluation and no comparison of alternative graph definitions	Same dataset and window setup (200 ms, 12 channels) as ours; its DB2 numbers are our main benchmark. We add the cross-subje

Sensors J.)			nd, 9 gestures, 3 arm positions)		scale temporal convolution (best at 7 layers)					ct test it leaves open
Graph Neural Networks for HD EMG-based Movement Intention Recognition: An Initial Investigation (IEEE RASSE)	S. M. Massa, D. Riboni, K. Nazarpou	2022	Public HD-sEMG database: 2 x 64 channels, 20 subjects, 65 isometric gestures, 2048 Hz	Amputee movement intention recognition; robotic prosthesis control	Graph with 128 nodes (channels) and 884 distance-based edges; node feature = 32 ms window (65 samples); GraphSAGE (6 layers) + mean pooling + MLP	Classification error rate. Mean 8.75% (std 4.9); 45 of 65 gestures under 10% error	First GNN on HD-EMG; very short 32 ms window (low latency); many gestures	No comparison with other ML/DL methods; per-subject training only; weak on combined multi-DoF gestures; simple heuristic graph	No baselines, no cross-subject test, graph chosen heuristically	Motivates a GNN on raw sEMG windows. We add baselines (CNN/LSTM) and a systematic graph comparison
A Graph Neural Network Model for Real-Time	P. Vipulanandan, K. Premaratne, M. Murthi	2026	Public Myo-band dataset (from De Silva et al.): 8	Real-time gesture detection for prostheses and AR	Graph per 0.4 s window: nodes = channels, edges	Accuracy, confusion matrix, ROC/AUC. Average	Very fast; includes onset detection; correlation graph	Only 5 gestures and 8 subjects; random split	Easy task and optimistic protocol; no cross-subject	Supports the correlation-graph option and real-time

Gesture Recognition Based on sEMG Signals (arXiv:2607.07850, preprint)			subjects, 5 gestures, 8 channels, 200 Hz, 20 repetitions per gesture		= Pearson correlation (fully connected, weighted); event-detection threshold to find gesture onset; simple GNN (2 fully connected layers, ReLU)	98.19% (90% overlap), 99.89% (97.5% overlap); 48 ms per graph +prediction on an M1 Pro CPU	needs no layout knowledge; robust to less training data (AUC 0.967 at 70% overlap)	over 90-97.5% overlapping windows (likely leakage, my reading); subject-specific; very simple GNN; preprint, not peer reviewed	test; no comparison with strong deep baselines	angle. We use a harder task (DB2, 49 gestures) and a leakage-free spli